work. Jupiter's Great Red Spot, a raging anticyclone that has been going strong for at least 350 years, is driven by identical physical processes that generate storms on Earth and elsewhere in the solar system.

Another class of universal truths is the conservation laws, where the amount of some measured quantity remains unchanged *no matter what*. The three most important are the conservation of mass and energy, the conservation of linear and angular momentum, and the conservation of electric charge. These laws are in evidence on Earth, and everywhere we have thought to look—from the domain of particle physics to the large-scale structure of the universe.

In spite of this boasting, all is not perfect in paradise. It happens that we cannot see, touch, or taste the source of eighty-five percent of the gravity we measure in the universe. This mysterious *dark matter*, which remains undetected except for its gravitational pull on matter we see, may be composed of exotic particles that we have yet to discover or identify. A small minority of astrophysicists, however, are unconvinced and have suggested that there is no dark matter—you just need to modify Newton's law of gravity. Simply add a few components to the equations and all will be well.

Perhaps one day we will learn that Newton's gravity indeed requires adjustment. That'll be okay. It has happened once before. Einstein's 1916 general theory of relativity expanded on the principles of Newton's gravity in a way that also applied to objects of extremely high mass. Newton's law of gravity breaks down in this expanded realm, which was unknown to him. The lesson here is that our confidence flows through the range of conditions over which a law has been tested and verified. The broader that range, the more potent and powerful the law becomes in describing the cosmos. For ordinary household gravity, Newton's law works just fine. It got us to the Moon and returned us safely to Earth in 1969. For black holes and the large-scale structure of the universe, we need general relativity. And if you insert low mass and low speeds into Einstein's equations they literally (or, rather, mathematically) become Newton's equations—all good reasons to develop confidence in our understanding of all we claim to understand.

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To the scientist, the universality of physical laws makes the cosmos a marvelously simple place. By comparison, human nature—the psychologist's domain—is infinitely more daunting. In America, local